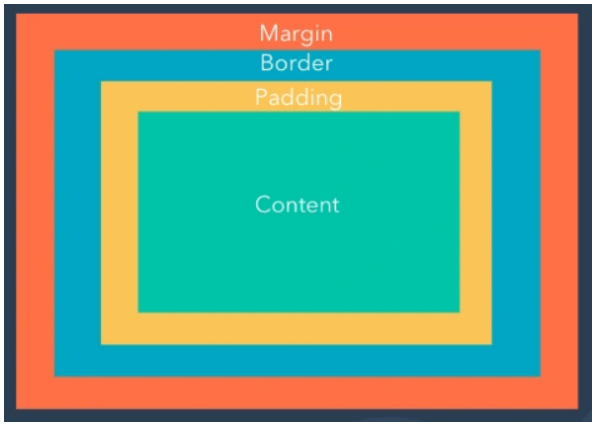
CSS

Day 5

Box Model:

- In CSS, every element on a webpage is represented as a rectangular box.
- This box is built using the box model, which describes how the size of an element is calculated and how it interacts with other elements.
- It consists of 4 main parts:
- Content → Padding → Border → Margin



Parts of Box Model

Content:

The actual text, image, or element inside the box.

Controlled by:

- width, height, min-width, max-width, min-height, max-height.

Padding

- Space between the content and border.
- Increases element size but background color extends into padding.
- Controlled by:
  - padding-top, padding-right, padding-bottom, padding-left.
  - Shorthand: padding: 10px 20px 15px 5px;

Border:

A line that goes around the padding and content.

Controlled by:

- border-width, border-style, border-color
- Shorthand: border: 2px solid black;
- Border properties are border radius

Margin

- Space outside the border, creates distance between elements.
- Background color does not extend into margin.
- Controlled by:
  - margin-top, margin-right, margin-bottom, margin-left.

- Shorthand: margin: 10px 15px 20px 25px;

Note:

By default element have some margin

margin: auto; only applies on left and right directions only not top and bottom

Examples:

Padding:

```
h3{
    border: 2px solid red;
}
```

Box Model 📦

Namaste css

```
h3{
    border: 2px solid red;
    padding: 10px;
}
```

Namaste css

```
h3{
    border: 2px solid red;
    padding-top: 10px;
}
```

Namaste css

```
h3{
    border: 2px solid red;
    padding-right: 10px;
}
```

Namaste css

```
h3{
    border: 2px solid red;
    padding-bottom: 10px;
}
```

Namaste css

```
h3{
    border: 2px solid red;
    padding-left: 10px;
}
```

Namaste css

Border:

Short hand property:

```
div{
    width: 50px;
    height:50px;
    border: 3px solid red;
}
```

```
div{
    width: 50px;
    height:50px;
    border-width: 5px;
}
```

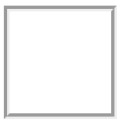
```
div{
    width: 50px;
    height:50px;
    border-width: 5px;
    border-style: solid;
}
```

```
div{
    width: 50px;
    height:50px;
    border-width: 3px;
    border-style: dotted;
}
```

```
div{
    width: 50px;
    height:50px;
    border-width: 3px;
    border-style: double;
}
```

```
div{
    width: 50px;
    height:50px;
    border-width: 3px;
    border-style: dotted;
}
```

```
div{
    width: 50px;
    height:50px;
    border-width: 3px;
    border-style: groove;
}
```



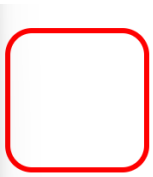
```
div{
    width: 50px;
    height:50px;
    border-width: 3px;
    border-style: ridge;
}
```

```
div{
    width: 50px;
    height:50px;
    border-width: 3px;
    border-style: inset;
}
```

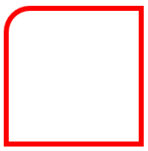
```
div{
    width: 50px;
    height:50px;
    border-width: 3px;
    border-style: outset;
}
```

```
div{
    width: 50px;
    height:50px;
    border-width: 3px;
    border-style: none;
}
```

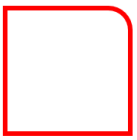
```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    border-radius: 10px;
}
```



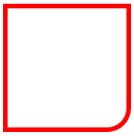
```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    border-top-left-radius: 10px;
}
```



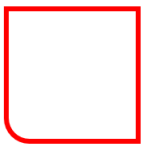
```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    border-top-right-radius: 10px;
}
```



```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    border-bottom-right-radius: 10px
}
```

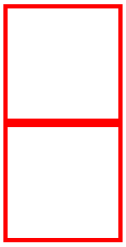


```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    border-bottom-left-radius: 10px;
}
```



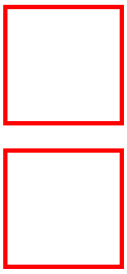
Margin:

```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    margin: auto;
}
```

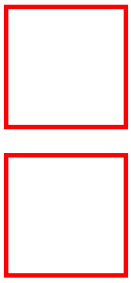


Short hand property:

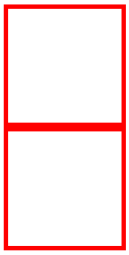
```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    margin: 10px;
}
```



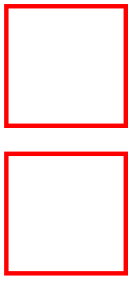
```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    margin-top: 10px;
}
```



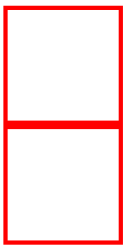
```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    margin-right: 10px;
}
```



```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    margin-bottom: 10px;
}
```



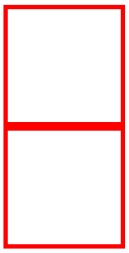
```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    margin-left: 10px;
}
```



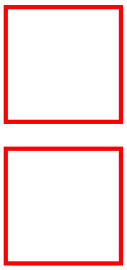
Note: margin:auto; works only in left and right not top and bottom.

Margin:auto; becomes the element in to centre as horizontally.

```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    margin: auto 10px;
}
```



```
div{
    width: 50px;
    height:50px;
    border: 2px solid red;
    margin: 10px auto;
}
```



Box-sizing:border-box VS Box-sizing:Content-box

Box-sizing:border-box;

Padding and border are included inside given width and height

Box-sizing:Content-box:

Padding and border are added outside of the given height and width

```
div{
    width: 100px;
    height:100px;
    border: 2px solid red;

}
```

Valibasha

```
div{
    width: 100px;
    height:100px;
    border: 2px solid red;
    padding: 20px;
    box-sizing: border-box;
}
```

Valibasha

```
div{
    width: 100px;
    height:100px;
    border: 2px solid red;
    padding: 20px;
    box-sizing: content-box;
}
```

Valibasha

Note:

Total width of an element = content width + padding left + padding right + border left + border right + margin left + margin right

Total height of an element = content height + padding top + padding bottom + border top + border bottom + margin top + margin bottom